The shortest day of the year in Northern hemisphere occurs during Winter solstice on 21/22 December. Contrary to it on 20/21 June longest day occurs in Northern hemisphere during the Summer solstice. Opposite to it, in Southern hemisphere, the shortest day occurs on 20/21 June and longest day on 21/22 December.

2. The Longest day in southern Hemisphere is

(a) 22 June (b) 22 December

(c) 21 March (d) 22 September

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

See the explanation of the above question.

3. In which hemisphere, roaring forties, furious fifties and shrieking sixties are blowing?

(a) Northern Hemisphere (b) Southern Hemisphere

(c) Eastern Hemisphere (d) Western Hemisphere

(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (b)

Roaring forties, furious fifties and shrieking sixties winds flow in the Southern hemisphere of the earth. They were given these names on the basis of the extensity of the Westerly winds nearby these latitude in the Southern hemisphere.

v. The Geological History

1. The southern continent broken from Pangaea is called

(a) Pacific Ocean (b) Laurasia

(c) Gondwanaland (d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Alfred Wegener a German Meteorologist was from "the continental drift theory" in 1912. This was regarding the distribution of the oceans and the continents. According to wegener, all the continents formed a single continental mass and mega ocean surrounded the same. The supercontinent was named PANGAEA, which meant all earth. The Mega Ocean was called PANTHALASSA, meaning all water. He argued that, around 200 million year ago, the super continent, Pangaea, began to split, Pangaea first broken into two large continents masses as Laurasia and Gondwanaland forming the northern and Southern components respectively. Laurasia and Gondwanaland continued to break into various smaller continents that exist today.

2. Continents have drifted apart because of-

(a) Volcanic eruptions

(b) Tectonic activities

(c) Folding and faulting of rocks

(d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Continents rest on massive slabs of rocks called tectonic plates, these plates are always moving and interacting in a process called Plate Tectonic Activity. Through these tectonic activities, continents shift position on Earth's Lithosphere.

3. Cocos Plate the between:

(a) Central America and Pacific Plate

(b) South America and Pacific Plate

(c) Red Sea and Persian Gulf

(d) Asiatic Plate and Pacific Plate

(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

In Carboniferous period, the whole land mass on the Earth was unified and was named as Pangea. This Pangea Supercontinent was surrounded by a superocean which was named 'Panthalassa' by Wegener. Due to the process of continental drift, the Pangea split into a northern most and southern most supercontinent. The northernmost landmass was known as Laurasia. The southernmost supercontinent was known as Gondwana land which split into South-America, Australia, Africa, Madagascar, Antarctica and Peninsular India. Cocos plate is a small lithospheric plate which lies between Central America and Pacific plate.

4. One astronomical unit is the average distance between

(a) the Earth and the Sun

(b) the Earth and the Moon

(c) the Jupiter and the Sun

(d) the Pluto and the Sun

(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (a)

An Astronomical unit (AU) is the average distance between Earth and the Sun, which is about 93 million miles or 150 million kilometer. Astronomical units are usually used to measure distance within our Solar System.

5. 'Rust Bowl' of the USA is associated with which one of the following regions?

(a) Great Lakes region (b) Alabama region

(c) California region (d) Pittsburg region
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Rust Bowl or Rust belt, a geographic region that was formerly a manufacturing or industrial powerhouse but now in deep decline phase as manufacturing hubs were moved overseas, the Great lake region and Pittsburg region comes under it.

6. Folding is the result of-

(a) Epeirogenic force (b) Coriolis force
(c) Orogenic force (d) Exogenic force

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The Earth's surface experiences different types of forces. The orogenic force takes millions of years to build a mountain from plain and sea bed. These forces due to the interaction between tectonic plates crumpled and pushed upward to form mountain range. Thus folding is the result of orogenic force.

7. An effective Coriolis force results in

(a) Solar System
(b) Earth rotation
(c) Interior of the earth
(d) Colorado and Gulf Stream
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

The direction of the winds blowing on the surface is determined by the air pressure and the rotation of the earth. The deflection in the direction of the winds is due to the repulsive force generated by the axial motion of the earth. Due to the discovery of this force by G.G. Coriolis, it was named Coriolis force. Thus one of the given options is the cause of the effective Coriolis force due to the rotation of the earth.

8. Which of the following statements is true about troposphere?

(a) Its average height is 13 km.
(b) It is the topmost layer of the atmosphere.
(c) The temperature at this layer increases with the height.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

The height of the troposphere varies from place to place and from season to season the height of the trophosphere at the pole is about 6–8 km. while at the equator it is about 16–18 km. its average height is approximetly 13 km.

9. Ocean thermal energy is produced due to

(a) pressure difference at different levels in the ocean
(b) temperature difference at different levels in the ocean
(c) energy stored in waves of the ocean
(d) tides rising out of the ocean
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Ocean thermal energy is produced due to temperature difference at different levels is the ocean. Energy from the sun heats the surface water of the ocean. In tropical regions, surface water can be much warmer than deep water. This temperature difference can be used to produce electricity and to desalinate ocean water. This whole process is known as Ocean Thermal Energy Conversion (OTEC).

10. The theory that states "pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle" is called

(a) the theory of continental drift
(b) the theory of Pangaea
(c) the theory of plate tectonics
(d) the theory of plate boundaries

69th B.P.S.C. (Pre) 2023

Ans. (c)

The theory of plate tectonics states-pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle. According to this theory, the earth surface is composed of several plates. There are seven major plates– Eurasian, African, Indo-Australian, Pacific, North American, South American and Antarctic plates. Significantly plates constitute the entire surface of the earth.

11. The process that continually adds new crust is

(a) subduction
(b) earthquake
(c) seafloor spreading
(d) convection

69th B.P.S.C. (Pre) 2023

Ans. (c)

The process that continually add new crust is seafloor spreading. Significantly, new crust is created by seafloor spreading. Old crust is destroyed by subduction. Seafloors spreading are only an aspect of plate tectonic, another is subduction.